

Balancing Collaboration and Discipline in Software Development Processes

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ABSTRACT

Neither traditional, agile or free/open software development models can be effective to all projects contexts. We claim that collaboration and discipline can be the driver to tailor software development processes to meet projects and organizations needs. This work proposes that process tailoring can be conducted through a context management approach.

Categories and Subject Descriptors

D.2.9 [Software Engineering]: Management – *software process models*.

General Terms

Management

Keywords

Software development processes, process tailoring, free/open source software, agile methods, collaboration, context management.

1. INTRODUCTION

Traditional, agile and free/open source software (FOSS) development models have the same goal: to improve software development, but they adopt different approaches. Traditional development, which is typically exemplified by maturity models, such as CMMI, seeks for predictability, stability and reliability [6], and is characterized by its orientation to planning and the emphasis on well-defined processes. Agile development, observed in Extreme Programming and Scrum, tries to quickly add value to business and adapt to changes [3]. In FOSS, the main objective is to guarantee users' freedom, working according to the bazaar metaphor [10], where projects are collaboratively developed. Although a lot of variation inside these categories exists, in general, each of them has important characteristics and has impacted software development.

Software organizations engage in a wide variety of projects with different characteristics, where traditional, agile and FOSS models are usually perceived as alternatives, but none of them will fulfill all requirements of a specific project or organization. Approaches that balance the underlying concepts in these different development models are necessary [4]. This need can be observed through the growth of processes tailoring by organizations, while decreasing the number of organizations that follow a reference model in a completely prescriptive manner.

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In this scenario, the research question investigated in this work is: how to tailor development processes, according to projects and organizations needs (context) considering collaboration and discipline as main drivers. The expected benefits are: increase the productivity of tailoring processes (reduction of the effort required to perform it), enhancing the quality and adequacy of processes generated, and reduce the risk of the resulted process not being adequate and used in practice.

2. RELATED WORK

According to the results obtained through a systematic review [7], several researchers have investigated the possibility of reconciliation among traditional, agile, and FOSS models. In general, the existing proposals involve the comparison and combination of the practices suggested by different models, aiming to produce a new hybrid one.

However, the complexity of software development and the variety of existing methods make the task of comparing them, arduous and inaccurate. This kind of software development models combination limits the potential for synergy among them, possibly resulting in an incomplete method, where it is no longer possible to ensure that the resulting process actually has the desired characteristics.

Boehm and Turner's proposal [4] suggest the risk analysis of the project characteristics as a way to select the project adequate method. This proposal has similarities with this research work, but it focuses only on agile and traditional, without considering FOSS development. We also changed the focus from agility to collaboration aiming to go beyond the reconciliation of models.

3. COLLABORATION AND DISCIPLINE

Discipline refers to plan and direct process during design phase, while collaboration focuses on people interaction during process execution. Both are complementary and essential in any project, but in different proportions, depending on project characteristics.

For a balanced mix between collaboration and discipline, it is necessary to understand how these aspects vary and distinguish the software development models. Regarding collaboration, we can consider the different levels of formality in communication, coordination, awareness and memory [8]. Regarding discipline, software models vary on emphasis and form of their processes. The traditional development is characterized by an emphasis on well-defined and continuously improved processes. Both agile and free/open development use no description or explicit modeling of the process adopted. Instead, they deal with a set of general principles to guide the development.

In order to explicit the collaboration and discipline, some instruments can be considered. To understand the **collaboration** existing among **people** in software projects, **social networks** [2]

appears as a promising path. A social network consists of a finite set of actors and the relationships among them. We can find several works on social network visualization and analysis, that points to social networks potential to explain how the collaboration occurs within a group and can help to calibrate the level of collaboration desired [11]. The **discipline** can be measured by regulating the level of control and formalism of **processes**. **Process lines** [12] represent an important instrument due to its flexibility to support process tailoring and the regulation of the level of formalism. A process line can be understood as a set of processes in a particular domain or for a particular purpose, having common characteristics and built based upon common, reusable process assets [12].

Collaboration and discipline vary during all software development lifecycle. It is impossible to predict, as project starts, a process appropriate to all situations. Therefore, process tailoring must be continuously executed, considering the process enactment context.

4. PROPOSED SOLUTION

The proposed solution is to tailor software processes, considering projects and organizations specific needs for collaboration and discipline, through the use of a context management approach. Context is a complex description of the knowledge shared on physical, social, historical and other circumstances where actions or events happen in the real world [5].

Two scenarios of context information use were selected. The first scenario is the activity of a project manager in defining a process for a new project. Scenario 2 consists of the activity of monitoring the project progress. These scenarios were chosen, based on literature and author's experience, because they represent important moments, where project manager must be provided with information to make decisions about the project future.

During scenarios execution, it is intended to offer an integrated visualization of similar contexts, of the process customized to the project, of the process actually executed, and of project social networks. These visualizations will help to increase project team awareness about the current state of collaboration and discipline.

A computing infrastructure is being designed to support these ideas. In this infrastructure, context information may be gathered from different data sources and combined to tailor a software development process that balances collaboration and discipline to the development project. During project execution other context information would be generated and captured, thus collaboration and discipline balance could be continuously reviewed and updated.

5. RESEARCH METHODOLOGY

As a starting point to establish relevant context information for software domain, Araujo et al. [1] suggest nine dimensions. We are particularly interested in organization, project and team dimensions. The former two dimensions are especially important considering that processes tailoring are realized in these two levels [9], and the team dimension is also relevant to our focus on collaboration.

For each of these dimensions, it is possible to think about a set of context information, but to determine which is relevant may not be a trivial activity. Some examples of context information are: organizational structure, organizational culture, knowledge management, problem size, client relationship, requirements stability, team experience, group working experience and team proximity. This initial set of context information has been obtained from literature review and author's experience.

The research methodology of this work also includes the construct

of a software process line, considering the development models. In addition, it is also important to investigate alternatives to implement the balance between collaboration and discipline.

6. CONCLUSION

The first result achieved in this research work was the characterization of software development models reconciliation, by conducting a systematic review of the literature. Another result obtained was the investigation of context representation models. The initial set of context information relevant to software development has been obtained and we intend to expand and validate this list through a survey with specialists (in academy and industry). In addition, the approach to link process line and social networks through context management has been detailed. Finally, the guidelines to identify the collaboration patterns in social networks were established [11].

Currently, this work is in the phase of designing the architecture of the proposed infrastructure. It includes: a social network module to offer collaboration analysis; a process line module to process line tailoring and visualization; a context management module composed by mechanisms to capture, store, retrieve, and visualize the context; and the balance module with the reasoning, to indicate the adequate mix of collaboration and discipline.

Finally, all this solution should be validated to verify: the possibility to obtain and present the context information; if context information will be helpful in process tailoring; and if resulted process contributes to project success. This validation includes the conduction of some experimental studies, like case studies.

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